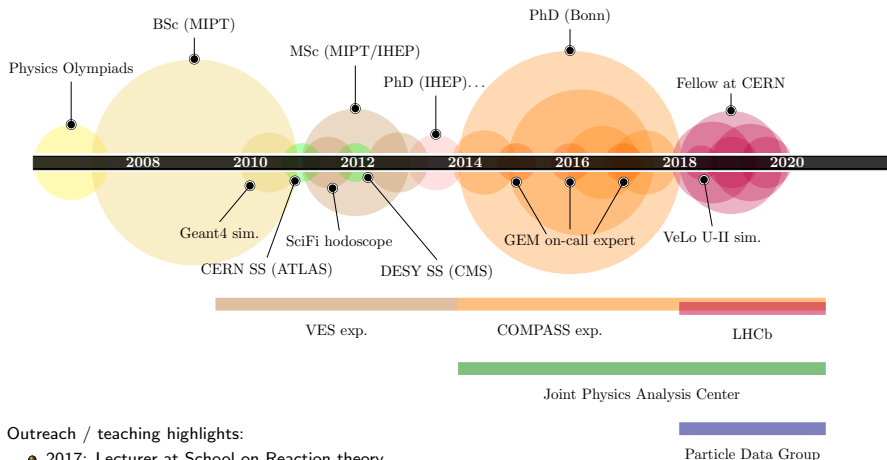


# Mikhail Mikhasenko: career path and projects



## Outreach / teaching highlights:

- 2017: Lecturer at School on Reaction theory
- CERN learning initiative (SS 2020): workshop on amplitude analysis
- COMPASS, LHCb guide

# Search for exotic hadrons and hadronic parameters

The need for the Amplitude Analysis

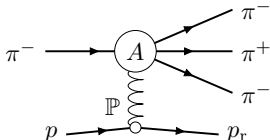
Understanding QCD at low energy: resonance physics, matter formation, input to CPV

# Search for exotic hadrons and hadronic parameters

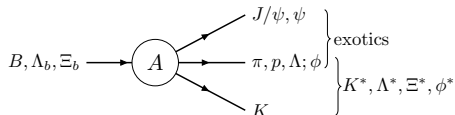
The need for the Amplitude Analysis

Understanding QCD at low energy: resonance physics, matter formation, input to CPV

**In scattering (COMPASS/VES):**



**In decays (LHCb):**

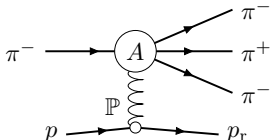


# Search for exotic hadrons and hadronic parameters

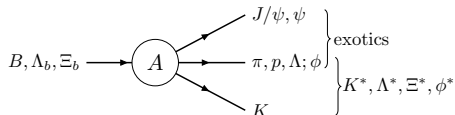
The need for the Amplitude Analysis

Understanding QCD at low energy: resonance physics, matter formation, input to CPV

**In scattering (COMPASS/VES):**



**In decays (LHCb):**



- Describe structures in the spectrum
  - ▶ identify conventional states
  - ▶ kinematic effects
  - ▶ distinguish from the cross-channel interaction

⇒ Amplitude Analysis

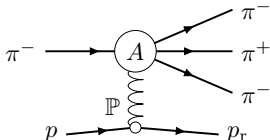
My contribution is filling the gap  
between theory and experiment.

# Search for exotic hadrons and hadronic parameters

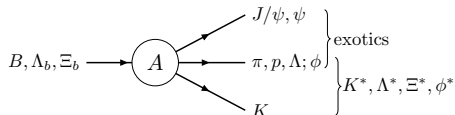
The need for the Amplitude Analysis

Understanding QCD at low energy: resonance physics, matter formation, input to CPV

**In scattering (COMPASS/VES):**



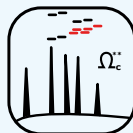
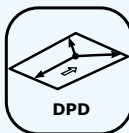
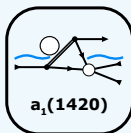
**In decays (LHCb):**



- Describe structures in the spectrum
  - identify conventional states
  - kinematic effects
  - distinguish from the cross-channel interaction

⇒ Amplitude Analysis

**amplitude analysis projects**



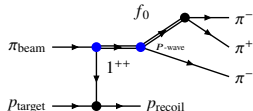
**hardware: SiPM shaper, ...**

**future plans: QTI, ...**

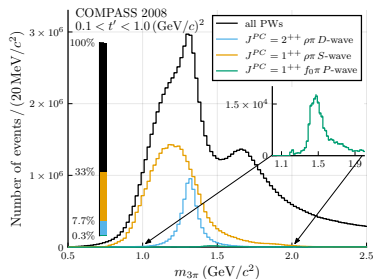
My contribution is filling the gap between theory and experiment.

# Light hadron exotics: the $a_1(1420)$

[(COMPASS) PRL 115 (2015) 082001]

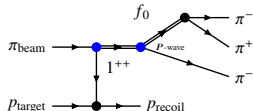


- COMPASS@2008: model-ind. partial wave analysis
- $100 \times 11$  fits,  $\sim 150$  parameters
- **A resonance-like behavior in  $f_0\pi$   $P$ -wave**

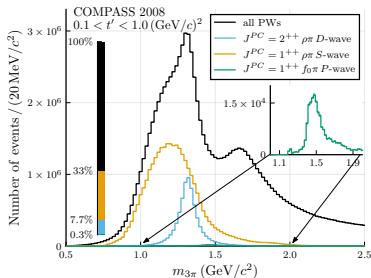


# Light hadron exotics: the $a_1(1420)$

[(COMPASS) PRL 115 (2015) 082001]



- COMPASS@2008: model-ind. partial wave analysis
- $100 \times 11$  fits,  $\sim 150$  parameters
- **A resonance-like behavior in  $f_0\pi$   $P$ -wave**

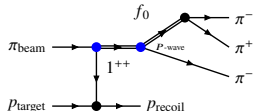


## My contribution:

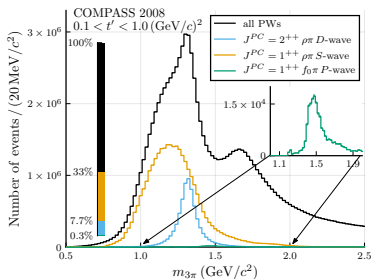
- syst:  $\pi\pi$   $S$ -wave parametrization
- stat: bootstrap using Julia [\[julialang.org\]](https://julialang.org)
  - ▶ ForwardDiff derivatives
  - ▶ parallelization ( $\sim 10^4$  core-hour)
- plotting
- interpretation (next slide)

# Light hadron exotics: the $a_1(1420)$

[(COMPASS) PRL 115 (2015) 082001]



- COMPASS@2008: model-ind. partial wave analysis
- $100 \times 11$  fits,  $\sim 150$  parameters
- **A resonance-like behavior in  $f_0\pi$   $P$ -wave**



## My contribution:

- syst:  $\pi\pi$   $S$ -wave parametrization
- stat: bootstrap using Julia [\[julialang.org\]](http://julialang.org)
  - ▶ ForwardDiff derivatives
  - ▶ parallelization ( $\sim 10^4$  core-hour)
- plotting
- interpretation (next slide)

## $a_1(1420)$ is not something ordinary:

- The mass is close to ground state  $a_1(1260)$
- Its width is narrower than the  $a_1(1260)$
- Mass is very close to the  $K^*\bar{K}$  threshold (!!)

New particle may be made of four quarks



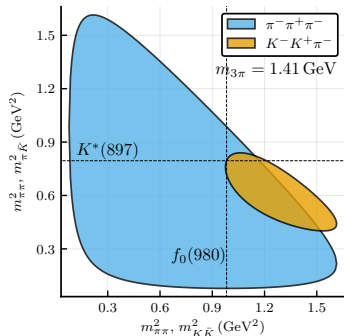


# Triangle singularity as a nature of the $a_1(1420)$ [MM, A. Sarantsev, B. Ketzner, PRD 91 (2015)]

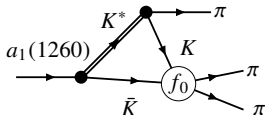
A three-body rescattering effect due to the coupling  $3\pi/K\bar{K}\pi$

- The rescattering  $K^*K \rightarrow f_0\pi$  kinematically allowed with a real  $K$ -exchange.

⇒ Landau singularity (TS)



- Inelasticity  $\pi\pi \leftrightarrow K\bar{K}$  causes redistribution of events between final states  $3\pi, K\bar{K}\pi$ .



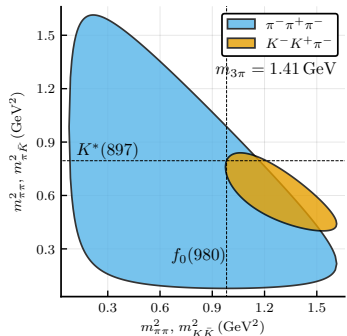
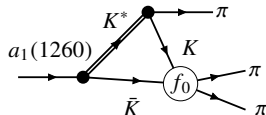
A theoretical idea has turned into to the first rigid observation of the TS

# Triangle singularity as a nature of the $a_1(1420)$ [MM, A. Sarantsev, B. Ketzer, PRD 91 (2015)]

A three-body rescattering effect due to the coupling  $3\pi/K\bar{K}\pi$

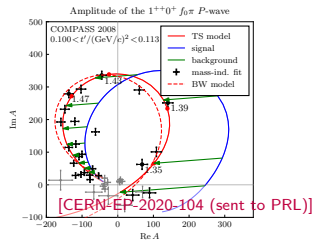
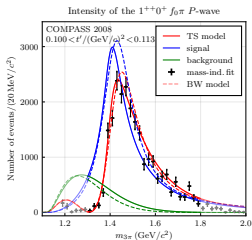
- The rescattering  $K^*K \rightarrow f_0\pi$  kinematically allowed with a real  $K$ -exchange.

⇒ Landau singularity (TS)



A theoretical idea has turned into to the first rigid observation of the TS

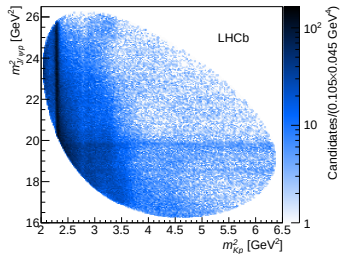
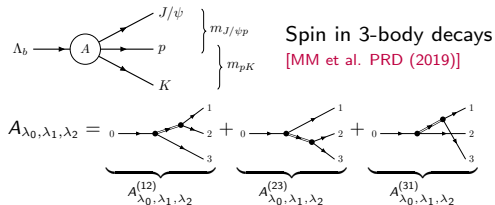
- Inelasticity  $\pi\pi \leftrightarrow K\bar{K}$  causes redistribution of events between final states  $3\pi, K\bar{K}\pi$ .



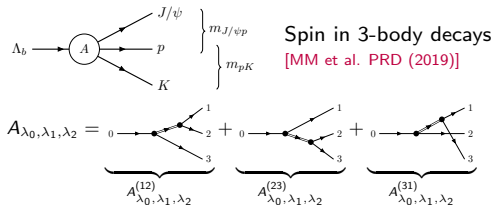
[CERN-EP-2020-104 (sent to PRL)]

- TS model has the same quality as the BW (dashed)
- No mass/width parameters: the shape is fixed
- The size of the effect matches the expectations

# Dalitz Plot Decomposition (DPD)



# Dalitz Plot Decomposition (DPD)



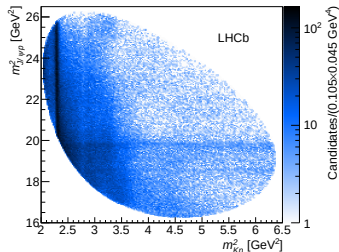
## Used in the past

$$A_{\lambda_0, \lambda_1, \lambda_2}^{(23)} = \underbrace{\boxed{\phi_i, \theta_i}}_{0 \rightarrow X, 1} \times \underbrace{\boxed{\phi'_i, \theta'_i}}_{X \rightarrow 2, 3} \times \underbrace{\boxed{\phi''_i, \theta''_i}}_{\text{spin align.}}$$

- "impossible" to get right
- spin 1/2:  $A(\pi) \neq A(-\pi)$
- range of  $\phi$  matters  $[-\pi, \pi]$  vs  $[0, 2\pi]$

## Consequences of the incorrect spin matching:

- Unpolarized decay is not inhomogenous
- Observed intensity is  $2\pi$  discontinuous in  $\phi$
- Broad components are impacted the most
- Determination of  $J^P$  might be incorrect



## Proposed in DPD

$$A_{\lambda_0, \lambda_1, \lambda_2} = \sum_{\nu} D_{\lambda_0, \nu}^{1/2*}(\alpha, \beta, \gamma) \underbrace{O_{\lambda_1, \lambda_2}^{\nu}(m_{12}^2, m_{23}^2)}_{O^{(12)} + O^{(23)} + O^{(31)}}$$

$$O_{\nu, \lambda_1, \lambda_2}^{(23)}(m_{12}^2, m_{23}^2) = \underbrace{\boxed{\phi_i, \theta_i}}_{0 \rightarrow X, 1} \times \underbrace{\boxed{\phi'_i, \theta'_i}}_{X \rightarrow 2, 3} \times \underbrace{\boxed{\phi''_i, \theta''_i}}_{\text{spin align.}}$$

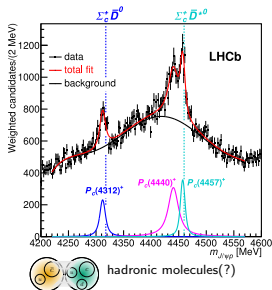
- correct by construction

## Impacted analyses:

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| $P_{cs}$               | $\Xi_b^- \rightarrow J/\psi \Lambda^0 K^-$ just published |
| $Z, X$                 | $B^- \rightarrow J/\psi \phi K^-$ at RC                   |
| $CPV$                  | $\Xi_b^- \rightarrow K^- p K^-$ at RC                     |
| $P_c$                  | $B_s^0 \rightarrow J/\psi p \bar{p}$ at RC                |
| $Z$                    | $B^0 \rightarrow J/\psi (\psi') \pi^+ K^-$                |
| $H_{\lambda, \lambda}$ | $\Lambda_c^+ \rightarrow p K^- \pi^+$                     |

# Better understanding of the resonance dynamics

**Lineshape** function is a key ingredient to learn underlying dynamics, and get a hint on microscopic picture.



$$\Lambda_b, \Lambda \rightarrow P_c, \mu \begin{matrix} J/\psi, \mu \\ p, \lambda \\ K \end{matrix} = \underbrace{\square}_{\Lambda_b^0 \rightarrow P_c, K^-} \times \underbrace{L(m^2)}_{\text{resonance dynamics}} \times \underbrace{\square}_{P_c \rightarrow p, J/\psi} \times \underbrace{\square}_{\text{spin align.}}$$

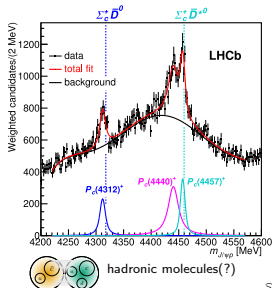
$$L(m^2) \stackrel{?}{=} \frac{1}{M^2 - m^2 - iM\Gamma} \quad (\text{BW approximation(!)})$$

I lead the effort on advanced methods:

- ⇒ Methods based on scattering theory (K, N/D)
- ⇒ Multichannel parametrization
- ⇒ Extraction of T-matrix pole parameters (+PDG)

# Better understanding of the resonance dynamics

**Lineshape** function is a key ingredient to learn underlying dynamics, and get a hint on microscopic picture.



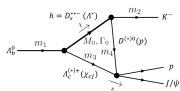
$$\Lambda_b, \Lambda \rightarrow P_c \rightarrow J/\psi, \mu \rightarrow p, \lambda \rightarrow K = \underbrace{\Lambda_b^0 \rightarrow P_c, K^-}_{\text{resonance dynamics}} \times \underbrace{L(m^2)}_{\text{resonance dynamics}} \times \underbrace{P_c \rightarrow p, J/\psi}_{\text{spin align.}}$$

$$L(m^2) \stackrel{?}{=} \frac{1}{M^2 - m^2 - iM\Gamma} \quad (\text{BW approximation(!)})$$

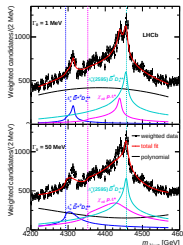
I lead the effort on advanced methods:

- ⇒ Methods based on scattering theory (K, N/D)
- ⇒ Multichannel parametrization
- ⇒ Extraction of T-matrix pole parameters (+PDG)

## Investigation of the TS



no acceptable description of  $P_c$  in the rescattering picture has been found



Impacted LHCb analyses:

$$P_c \quad \Lambda_b^0 \rightarrow J/\psi p K^- \text{ (publ.)}$$

$$Z, X \quad B^- \rightarrow J/\psi \phi K^- \text{ (in RC)}$$

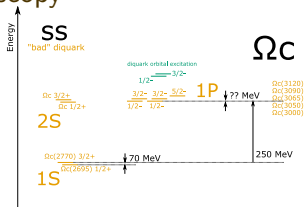
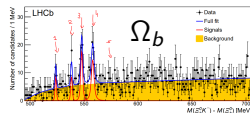
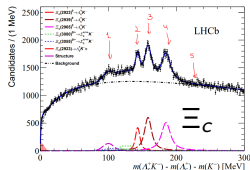
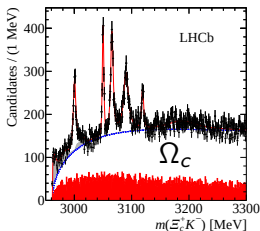
$$X(3872) \quad B^- \rightarrow J/\psi \pi^+ \pi^- K^-$$

$$\Lambda_b^{**} \quad \Lambda_b^{**0} \rightarrow \Lambda_b^0 \pi^+ \pi^- \text{ (publ.)}$$

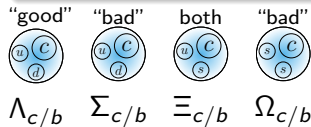
$$X/T_{cc} \quad D\bar{D} / D\bar{D}^* / DD / DD^*$$

# Heavy baryons spectroscopy

## Five lines mystery

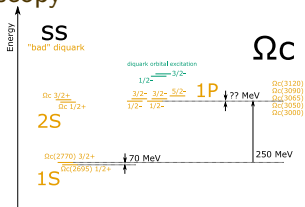
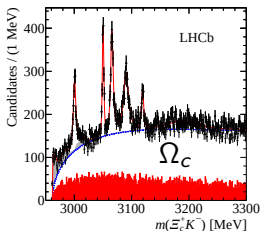


5 lines pattern is a feature of the "bad" diquark

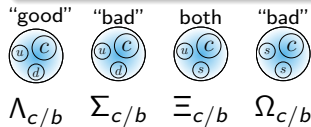


# Heavy baryons spectroscopy

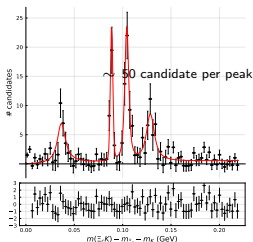
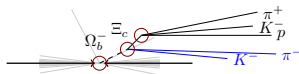
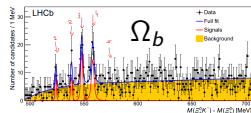
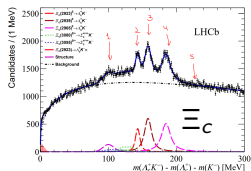
## Five lines mystery



5 lines pattern is a feature of the "bad" diquark



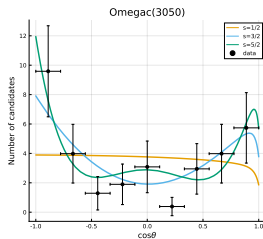
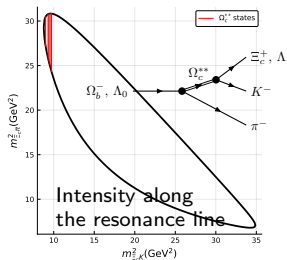
First observation of  $\Omega_c^{**}$  in the exclusive reaction



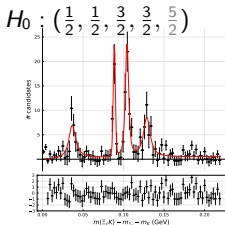
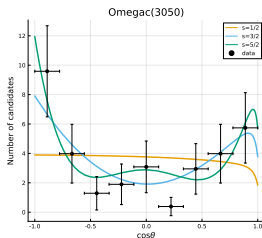
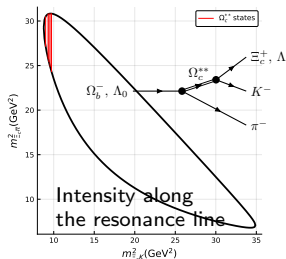
- Selection
- Kinematic fitting
- Efficiency map
- Resolution MC
- FeedException
- Significances
- Test Statistics



# $\Omega_c^{**}$ analysis



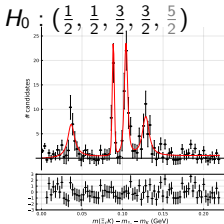
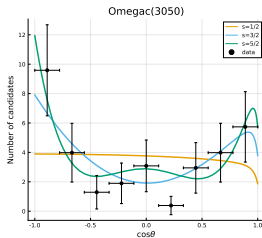
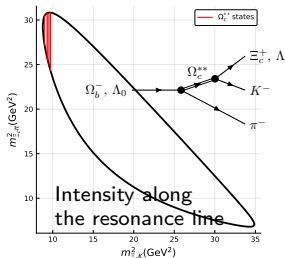
# $\Omega_c^{**}$ analysis



- $H_0$  is excl. with  $> 3\sigma$
- Instead,  $(\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{5}{2})$  is supported
- missed  $1/2$  state (?): either broad, or below the threshold, or ...

Analysis passed the WG review  
On track for Moriond QCD 2021

# $\Omega_c^{**}$ analysis



$J \neq 1/2$

- $H_0$  is excl. with  $> 3\sigma$
- Instead,  $(\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{5}{2})$  is supported
- missed  $1/2$  state (?): either broad, or below the threshold, or ...

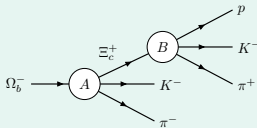
Analysis passed the WG review  
On track for Moriond QCD 2021

## “Polarization Sensitivity”:

(I just invented the quantity)

is a property of the decay expressing how the final state reflects polarization

- Property of hadron decay
- Can be measured once and transferred to any other decay
- Needs generalization for  $3b$  decay

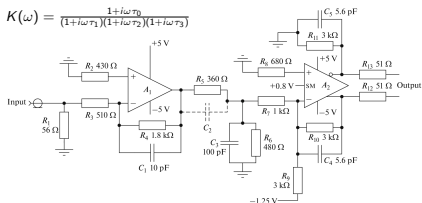


$$M = A(m^2, m^2)D(\alpha, \beta, \gamma)B(m^2, m^2)$$

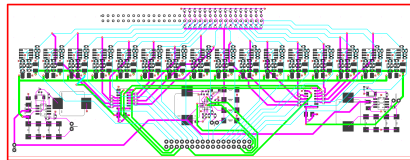
Analysis of  $\Xi_c \rightarrow pK\pi$   
CERN SS 2020 project

## Hardware and other experiences

$$K(\omega) = \frac{1 + j\omega\tau_0}{(1 + j\omega\tau_1)(1 + j\omega\tau_2)(1 + j\omega\tau_3)}$$



[Instrum. Exp. Tech. **63** (2020) no.2, 165-182]

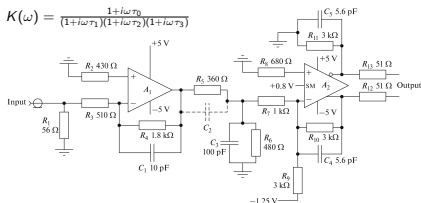


I have designed, traced, soldered, and tested the 16-channels shaper board.

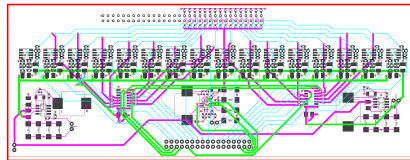
- tail suppression of SiPM signal, differential output to ADC
- used with SiPMs (SciFi hodoscope), adopted to PMT (calorimeter).

## Hardware and other experiences

$$K(\omega) = \frac{1+j\omega\tau_0}{(1+j\omega\tau_1)(1+j\omega\tau_2)(1+j\omega\tau_3)}$$



[Instrum. Exp. Tech. **63** (2020) no.2, 165-182]



I have designed, traced, soldered, and tested the 16-channels shaper board.

- tail suppression of SiPM signal, differential output to ADC
- used with SiPMs (SciFi hodoscope), adopted to PMT (calorimeter).

## Commissioning and operation

### GEM detectors at COMPASS

- Calibration and maintenance
- On-call duty during 2016-2018 runs

### CMS Pixel detector (DESY SS)

- Test beam measurements
- Modeling of the charge distribution

## Simulation projects

### U-II VeLo: 4d tracker simulations

[VeLo framework TDR (under WG review)]

### Geant4 simulations of the Active Target

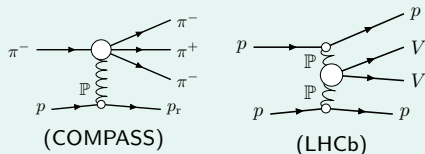
[my BSc thesis]

## Future plans

### CERN Quantum Technology Initiative: group of enthusiasts at CERN

- I have initiated a journal club on Th.&Comp.
- Plan to advise a PhD project exploring QC/ML

### From COMPASS diffraction to CEP



[MM et al., arXiv:2007.05501]

### “Polarization Sensitivity” for AmAn

- Establish the concept
- Measure  $\Lambda_c^+, \Xi_c^+ \rightarrow ph^+ h^-$
- Transfer the knowledge to complex decay chains, e.g.  $\Lambda_b^0 \rightarrow \Lambda_c^+ \bar{D}^0 K^-$

[CERN-STUDENTS-Note-2020-031 (advisor MM)]

### Spin and parity of exotics

$$P_c, P_{sc}, Z_c, Z_{sc}$$

### Angular analysis of $c/b$ baryons

$$\Lambda_{c/b}^{**}, \Sigma_{c/b}^{**}, \Xi_{c/b}^{**}, \Omega_{c/b}^{**}$$

### VeLo Upgrade II concept and detailed studies

Support for AMBER/COMPASS++ Phase-II (kaon, antiproton spectroscopy)

PDG reviews ( $a_1(1260)$ , revision of  $K^*$ , a big review on hadrons)

[MM et al., PRD 98 (2018)]